

When Test-Time Search Exploits Latent Costs in Robot World-Model Planning

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Abstract

Action-conditioned JEPA world models plan by coupling predicted latent futures to a representation-induced goal cost and a test-time optimizer. Published systems succeed at reaching and pose control, but remain brittle on pushing, grasping, and pick-and-place. We localize one failure at the optimizer–cost interface. In MetaWorld, an *oracle ladder* rolls every candidate through the true simulator and re-encodes it, making latent dynamics perfect while the cost is varied. The same planner solves push 16/16 with a true-state cost, whereas all frozen-encoder costs tested solve only 0–2/16, including a readout that decodes 91.5% of generic off-policy frames within the task’s 5 cm tolerance. The apparent paradox is selection-induced: that readout is within tolerance on only 24% of CEM elites, and simulator state confirms that the selected objects are genuinely far from goal. An informative true-state-cost control converts added search into success, whereas contact costs improve their proxy without improving the true outcome; the pressure trend is suggestive and the direct elite-selection gap is the primary evidence. Across four released DROID checkpoints from 22M to 1B parameters, an observational action-ranking diagnostic shows no upward trend, but it is not a controlled scaling law. Together the results show that average prediction, action-discrimination, and readout accuracy do not guarantee a cost that survives optimization. We therefore propose an elite-conditioned proxy–truth audit for world models intended for planning. The claim is deliberately scoped to the tested checkpoint, cost, task, and planner combinations; grounding alone is insufficient, and broader planner/checkpoint confirmation is evaluated separately.

1 Introduction

Action-conditioned world models that predict in a learned latent space — the JEPA family (DINO-WM (Zhou et al., 2024), V-JEPA-2-AC (Assran et al., 2025), and the design study of Terver et al. (2025)) — plan by coupling two learned components with a test-time optimizer: a predictor imagines futures, a cost scores them against a goal, and a sampling planner (CEM) searches for the action sequence that minimizes the cost. On free-space tasks this works well. But a consistent pattern runs through the field’s own evaluations: the *positive* closed-loop numbers are for reaching and pose control, while contact-rich tasks — pushing, grasping, pick-and-place — are omitted, demoted to qualitative figures, or propped up with hand-crafted sub-goals. Terver et al. (2025) stop their closed-loop Metaworld tables at Reach and note the model “hallucinates grasping the object”; Assran et al. (2025) report pick-and-place only after supplying three manual sub-goal images.

One common diagnosis is a *model-capacity* story: the encoder is too small, the representation too coarse, or the predictor too weak. We isolate a distinct failure that capacity and average prediction metrics do not capture: **the optimizer–cost interface**. A CEM planner at test time is an optimizer pitted against a learned proxy objective. The reinforcement-learning-from-human-feedback literature has a name for what happens to optimizers pitted against learned proxies: *overoptimization*, or Goodhart’s law — push on a proxy hard enough and the true objective decouples from it (Gao et al., 2023; Skalse et al., 2022; Karwowski et al., 2024; Amodei et al., 2016). We show the same phenomenon can occur in latent world-model planning, with one operational difference: in RLHF the optimizer is a training loop one can regularize; here the optimizer runs *at every control step* and selects candidates precisely because the cost scores them favorably.

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This diagnosis is not new; the isolation and the survival test are. A concurrent cluster of work, built mostly on small, non-robot latent world models, converges on a compatible reading — predictive accuracy does not imply a plannable cost. We do not claim first discovery that cost geometry, not the predictor, is the bottleneck (Sec. 2 compares each account in detail). What we add is an intervention: an oracle ladder that excludes learned-dynamics error by holding dynamics perfect and varying only the cost, rather than inferring it from an aggregate success gain, and a mined-elite test for whether a repaired cost *survives* being searched against by the same optimizer that broke it. The observational diagnostic spans real-robot DROID checkpoints up to 1B parameters; the adversarial survival test itself uses MetaWorld, where simulator state provides an independent verifier.

Three features distinguish our account from the classic “model exploitation” concern in model-based RL (Janner et al., 2019; Lambert et al., 2020), where a policy exploits errors in learned *dynamics*. (i) We give the planner *perfect dynamics* — every candidate action sequence is rolled in the simulator and re-encoded by the real frozen encoder — and the failure persists. Learned-dynamics error is therefore excluded within this harness, localizing the tested failure to cost and search. (ii) Within the frozen-encoder regime, a ladder of increasingly strong cost repairs tested here (better formula, better readout, off-policy hardening, encoder fine-tuning, ensembling) all fail the same contact gate. (iii) We show the standard diagnostic quantities the field might use to catch this — prediction error, action-grounding scores, readout precision — are all satisfied while planning fails, i.e. the failure is invisible to current evaluation practice.

Concretely, our experimental chain (MetaWorld with simulator ground truth for isolation and verification; DROID as an observational real-robot transfer check) supports three contributions:

- **Observational diagnosis across released checkpoints (Sec. 3).** Against dataset-mined, nearby-state action alternatives, four frozen DROID checkpoints from 22M to 1B parameters show no improvement in factual-action ranking. MetaWorld exhibits the same pre-grasp-worst ordering. Because the alternatives originate from neighboring rather than identical states, this is a discriminability diagnostic, not a causal intervention or controlled scaling law; a snapshot/restore benchmark is the required validation.
- **Oracle localization and optimizer-induced selection (Secs. 4–6).** Our harness first matches the released Reach result within uncertainty and reproduces the contact wall. Holding dynamics simulator-perfect, the true-state cost solves push 16/16, while every frozen-encoder cost tested stays at 0–2/16. A hardened readout that is 91.5% accurate on random off-policy frames falls to 24% on CEM elites; simulator state independently confirms the selected outcomes are far from goal. An informative true-state control converts search into success.
- **An elite-conditioned cost audit (Sec. 8).** We separate average-case readout quality from selection-induced readout shift and simulator-verified outcomes on the states the optimizer actually trusts. This audit exposes exploited costs that prediction error, action-discrimination, and average readout accuracy miss. Hardened probes, representation adapters, encoder LoRA, and disagreement ensembles do not cross the contact wall; these are scoped nulls, not a universal impossibility result.

Conventions (how to read our numbers). Four kinds of quantities appear throughout, and we keep their formats distinct. *Planning success* is always episodes solved out of 16 paired evaluation episodes, written $k/16$ (with the percentage in parentheses when comparing to published baselines). *Readout accuracy* is always the percentage of frames whose decoded object position falls within the 5 cm task-success radius (“% within 5 cm”), with the median error in cm where informative. *CRA* is a ranking accuracy over transitions, reported as a fraction in tables; with 16 exchangeable distractors the nominal baseline is $1/17 \approx 0.059$, but mined neighbours need not be exchangeable. *AUG/ECS* are raw latent-space gaps (Sec. 3.1), meaningful only within a model. Distances in the workspace are in cm or in Metaworld state units where noted.

Planner conventions. We write a CEM budget as (samples)×(iterations) evaluated per replan — e.g. 100×6 is 100 candidate action sequences refined over 6 CEM iterations — with the rollout horizon H stated separately. At each iteration CEM keeps the top-scoring candidates, the *elites*, and refits its sampling distribution to them; the converged elite population is the set of futures the planner “trusts”. Pools marked “ n_{eff} -weighted” average per-regime cells weighted by their effective (trajectory-clustered) sample count, so a few long trajectories do not dominate.

The through-line is a reframing of what world-model evaluation must report. Prediction error is satisfied while planning fails (Sec. 3); grounding is satisfied while planning fails (Sec. 6); readout precision is satisfied while planning fails (Sec. 5). The direct mechanism signal is disagreement between what the cost claims and what simulator state verifies *on candidates selected by optimization*. We propose optimizer-conditioned audits as a first-class evaluation axis for world models intended for planning.

2 Related work

JEPA world models for planning. DINO-WM (Zhou et al., 2024) predicts forward in the frozen feature space of a self-supervised image encoder and plans with sampling-based MPC from a goal image; V-JEPA-2-AC (Assran et al., 2025) learns an action-conditioned world model on a video encoder and demonstrates zero-shot manipulation; Terver et al. (2025) systematically study the design choices that make such models plan. We take these as frozen baselines and treat their own evaluations as evidence: across all three, strong closed-loop results are pose/free-space tasks, and contact is where the story thins (hand-crafted sub-goals, qualitative figures, “hallucinated grasping”). V-JEPA 2.1 (Mur-Labadia et al., 2026) subsequently improves dense spatial features and reports a large real-robot grasping gain, making representation quality and observability an important alternative explanation. Our study does not rule that out; it asks whether the costs induced by the released planning checkpoints survive their optimizer’s own selection.

Overoptimization and reward hacking. Optimizing against a learned proxy eventually decouples proxy from truth: formalized for RLHF reward models by Gao et al. (2023), characterized in general by Skalse et al. (2022) and Karwowski et al. (2024), and anticipated as “reward hacking” (Amodei et al., 2016). Its *test-time* variant is now well documented for language models, where search or RL against a learned verifier converges onto the verifier’s error pockets (Helff et al., 2026). We port this lens to latent-space *planning*, where it has not been quantitatively established: the CEM inner loop is the optimizer, the latent cost is the proxy, and the elite population is the analogue of the over-optimized policy — with the aggravation that the optimizer re-runs at every control step. Our oracle ladder gives this port an isolation the RLHF setting lacks — we can hold dynamics *perfect* and vary only the cost.

Model exploitation in model-based RL. That policy optimization exploits errors in learned *dynamics* is a classic MBRL concern (Janner et al., 2019), running through the sampling-MPC lineage our CEM planner inherits (Chua et al., 2018; Nagabandi et al., 2019), related to the objective mismatch between likelihood training and control performance (Lambert et al., 2020), and addressed offline with pessimism (Yu et al., 2020b; Kidambi et al., 2020); learned agents more broadly admit adversarial exploitation of their blind spots (Gleave et al., 2020). Concurrently, Parthasarathy et al. (2025) show *gradient-based* planners exploit a world model’s input gradients and respond by smoothing the dynamics. Our finding is materially different from all of these: with dynamics oracle-controlled — the channel every one of those works attacks or repairs closed entirely — the *cost* is exploited at test time, and the standard dynamics-side remedies (ensembles, pessimism) do not touch it (Sec. 6).

Action-conditioned reliability diagnostics. ATM (Chen et al., 2026) probes action semantics in real and predicted latent transitions and uses action-identifiability supervision to improve downstream planning. UWM-JEPA (Radha and Goktas, 2026) shows in a physics setting that teacher-forced JEPA objectives admit action-invariant solutions which counterfactual targets can repair. WAV (Liu et al., 2026) decomposes prediction verification into state plausibility and action reachability, then uses an inverse model for self-improvement. MiraBench (Yang et al., 2026) evaluates action-following and optimism under failure-inducing robot actions, while What-If World (Cai et al., 2026) evaluates paired physical interventions on shared scenes. These works preclude a novelty claim for action-identifiability or counterfactual evaluation alone. Our distinct target is optimizer-induced selection of cost errors under oracle-controlled dynamics; the DROID ranking study is supporting context, not the main contribution. IMWM (Gao et al., 2026) observes that finite-budget sampling planners can be the bottleneck even under an idealized world model and responds with demonstration priors — we instead show that the same finite-budget planner which solves the task with a true-state cost fails with tested frozen-encoder costs under perfect dynamics (Sec. 5). Vision-tactile world

models (Zhang et al., 2026) address what information contact planning needs; our results say that adding information without addressing exploitability is not sufficient (Sec. 5).

Closest cost and planner-interface diagnoses. A cluster of concurrent work built on a small, non-robot latent world model (Maes et al., 2026) reaches a compatible diagnosis by different routes. RC-aux (Li et al., 2026b) makes the framing explicit — “predictive but not plannable”: accurate short-horizon prediction can still leave a latent whose Euclidean distance does not reflect finite-horizon reachability — and repairs it with a training-time reachability objective that reshapes the representation (nearer to our relearned-representation arm, which also fails to cross contact, than to a purely post-hoc metric). Trajectory reachability metrics (TRM) (Li et al., 2026a) instead add a post-hoc terminal-ranking head on the frozen backbone. TRM is the closest cost-side study: it deploys the repaired metric inside CEM, audits ordering on identical candidates, compares against true-state cost, intervenes on the task-relevant subspace, and stress-tests larger search budgets. ACID (Seo et al., 2026) argues, in the same spirit, that a terminal-only latent cost cannot see whether intermediate transitions are realizable, and adds an inverse-dynamics cycle-consistency term to the planning cost. IMWM (Gao et al., 2026) already replaces learned dynamics with literal environment rollout and finds that many failures arise because no successful candidate enters the finite CEM population. Thus neither latent-cost misranking nor oracle-dynamics failure alone is our novelty.

Our narrower target is a contact-rich robot regime in which we jointly vary dynamics and cost, inspect optimizer-selected elites in simulator object state, and separate candidate coverage from cost-based selection. Relative to IMWM, the necessary discriminator is whether physically better candidates are present but misranked; relative to TRM, it is whether its reachability metric survives fresh search on held-out MetaWorld episodes. Our mined-repair-re-mine arm is potentially distinct only when mining, repair training, and final evaluation are strictly disjoint, so we treat it as supporting evidence pending that provenance audit. Direct TRM and ACID comparisons are required for a broad mitigation claim.

Amortizing away the search. Closest to our “remove the adversary” control (Sec. 8), Nguyen et al. (2026) replace CEM entirely with a goal-conditioned inverse-dynamics map and argue the opposite thesis — that a well-regularized latent has “already encoded” the structure search would recover, so the planner is not the bottleneck. Their evidence carries our wedge: the single contact-rich cell in their toy suite (Push-T) is the one place amortization stops dominating, which they attribute to “delicate multi-step contact.” Our MetaWorld E1 similarly finds that the tested amortized controller does not move the object. It localizes a second failure outside search without deciding whether the cause is representation, controller training, or coverage.

Reconciling three diagnoses of the same planner. Three concurrent accounts of why latent planners stall appear to conflict: IMWM (Gao et al., 2026) finds that even under oracle dynamics a finite-budget sampling planner is *search-limited* and adds demonstration priors; Nguyen et al. (2026) argue the planner is *not* the bottleneck — a well-regularized latent has already encoded what search would recover, so amortization suffices; we argue the *cost* is exploited by the very search that is supposed to help. These are not contradictory once conditioned on the regime, and the unifying variable is whether the cost has worst-case pockets *in that regime*. Where the cost is informative (free space, pose control), more search converts to success (IMWM’s lever; our reach control, 5/16 $\rightarrow$ 16/16, Fig. 1) and amortization can match it (Nguyen et al., 2026); where the cost is exploitable (contact), *both* levers fail — adding search digs into the pockets (Sec. 6.2) and removing it leaves a representation that never moves the object (our E1, Sec. 8). The apparent disagreement is a regime confound: each prior account reads its benchmark’s dominant regime as the whole story. Our oracle ladder makes the conditioning explicit — same planner and budget throughout, only the cost’s exploitability varies — which is why we can locate the binding constraint at contact as neither search coverage (IMWM) nor search presence (latentgeom) but cost exploitability.

Evaluating world models for decision-making. A concurrent position paper argues world models should be evaluated by their decision-making utility rather than predictive accuracy, and proposes a broad utility taxonomy (Yu et al., 2026). We agree with the position; our contribution to it is a mechanism and an instrument the taxonomy lacks — evidence that the prediction/planning gap is *adversarial* (the

planner manufactures the states on which the cost is wrong), and a concrete, cheap measurement of it (the exploitation gap, Sec. 8) computed from frames the planner already produced.

Benchmarks. We evaluate on Metaworld (Yu et al., 2020a), whose simulator state lets us supervise and *verify* costs against ground truth, and DROID (Khazatsky et al., 2024) as the real-robot transfer check. Planning follows the upstream CEM configuration (Rubinstein, 1997).

3 Observational diagnosis: weak action discrimination near contact

We first measure how well frozen predictors discriminate the factual action from actions mined at nearby dataset states. These are observational, near-state queries: they do not provide the true successor of an alternative action from the identical simulator state. The diagnostic can localize a failure mode, but causal action-effect claims require the snapshot/restore intervention described in Sec. 8. Full definitions and the AUG/ECS corroboration are in App. B and App. F.

3.1 Metrics

Let $F(z, a)$ be the frozen one-step latent predictor (driven through the model’s own `unroll` primitive, with the model’s own action normalisation) and d the upstream planning distance (L_2 for every baseline, matching the shipped `L2_cem` configuration). We measure three complementary quantities on held-out factual transitions (z_t, a_t, z_{t+1}) ; each answers a different question.

CRA — does the model rank the executed action best? Given the transition, we sample $K=16$ distractor actions $\{a_k^-\}$ and ask whether the factual action’s prediction lands closer to the realised next latent than every distractor’s prediction:

$$\text{CRA} = \Pr \left[d(F(z_t, a_t), z_{t+1}) < \min_{k=1 \dots K} d(F(z_t, a_k^-), z_{t+1}) \right], \quad (1)$$

The value $1/(K+1) = 1/17 \approx 0.059$ is the nominal baseline only when factual and distractor actions are exchangeable; the dataset-mined sampler below is not exchangeable, so an empirical action-ignoring null is required for a calibrated chance claim. The distractor family controls what the metric tests: *opposite* distractors ($a^- = -a_t$) test coarse directional grounding — the quantitative version of the “open vs. close the gripper” visualisations in prior work. For the sampler called *hard-NN*, we first retain candidate transitions within a latent radius (falling back to the K nearest states when the radius is under-filled) and then select the K actions farthest from the factual action. Thus it is a stress test of factual-action retrieval under a model-specific neighborhood, not an intervention and not automatically comparable across models. We report it because it motivated the oracle study, not as the paper’s primary causal evidence.

AUG — does the action help at all? The Action Usage Gap is the prediction-error increase when the action is replaced by another action from the batch (a permutation π):

$$\text{AUG} = \mathbb{E} \left[d(F(z_t, \pi(a_t)), z_{t+1}) - d(F(z_t, a_t), z_{t+1}) \right], \quad (2)$$

which is ≈ 0 when shuffling does not change average error. This is consistent with action-ignoring but is not an if-and-only-if test: signed differences may cancel and shuffled actions may be off-support. AUG is a raw latent-space gap: informative *within* a model, not comparable *across* models (latent scales differ), which is why it corroborates CRA rather than replacing it (App. F).

ECS — the same question, asked where it matters. Averaged over all transitions, AUG can be diluted by near-static steps (the arm hovering; the scene unchanged) where different actions genuinely lead to similar futures. Effect-Conditional Sensitivity restricts AUG — and, in the *effect-conditioned CRA* we report as the primary signal, the ranking — to transitions that actually move the latent, $\|z_{t+1} - z_t\| > \tau$ with τ the per-model median displacement. This removes many quiet transitions, but it also selects a different latent-defined subset for each model and can include ego-motion or nuisance variation. Cross-model claims therefore need a shared, model-independent physical-effect mask.

Boundary Blindness (BB; definition in App. B) is our fourth instrument. On a neighborhood of similar but non-identical states, it compares dispersion of observed outcomes with dispersion of model predictions under the neighbors’ actions, both z-scored over the population; $BB > 0$ indicates an observational sensitivity shortfall. Initial-state, policy, task, and hidden-contact differences can also increase the observed fan. On DROID the outcome itself is a $\|\Delta z\|$ proxy, not independent world ground truth.

Protocol. Each transition is assigned to one of four mutually exclusive regimes — free space, pre-grasp, gripper actuation, contact manipulation — by a fixed-threshold priority cascade on the simulator state: gripper actuation if the gripper aperture changes, else contact manipulation if the object moves (> 5 mm), else pre-grasp if the hand is within 10 cm of the object, else free space (full thresholds, and the DROID gripper-sensor and RoboCasa grasp-flag analogues, in App. A). This is an object-displacement *proxy*, not a MuJoCo contact sensor — the released data ships no contact flags — and the caveat is carried on every figure. All confidence intervals are trajectory-clustered bootstrap. Every frame is encoded once and cached; nothing is trained for the diagnostic.

3.2 Frozen baselines fail on nearby actions, at the boundary

Table 1: Effect-conditioned factual-action ranking accuracy (top-1, n_{eff} -weighted pool, Metaworld). Against *opposite* distractors the models are near-perfect — their predictions respond to coarse action changes. Against the hard-NN stress-test they score lower, worst at pre-grasp. Because hard-NN is near-state and action-maximizing, this table motivates rather than proves a same-state contact bifurcation.

model	distractor	free space	pre-grasp	contact
DINO-WM	opposite	0.990	0.961	0.975
	nearest-neighbour	0.566	0.467	0.481
JEPA-WM	opposite	0.993	0.972	0.984
	nearest-neighbour	0.635	0.541	0.571

The observational Boundary-Blindness score localizes the same pattern: pooled BB_{boundary} concentrates at the pre-grasp boundary on both model families (DINO-WM 1.323, JEPA-WM 1.280, vs. free-space 0.28–0.30; per-task CI-aware pairing elevated in 4/6 and 5/6 tasks with zero confident reversals) and transfers to real-robot DROID data (DINO-WM 1.916 [1.560, 2.279] under the $\|\Delta z\|$ proxy; Table 2 gives the full four-model breakdown). A conditional-mean account is compatible with this pattern when observations alias multimodal contact outcomes, but the observational neighborhood alone does not identify that mechanism.

3.3 No upward trend across four released checkpoints

We score four frozen DROID world models on an identical 333-episode subset: DINO-WM (DINOv2 ViT-S, 22M), JEPA-WM (DINOv3 ViT-L, 300M), V-JEPA-2-AC (ViT-G, 1.01B), and the independent public V-JEPA-2-AC-OSS (1B).

The model-native scores show no upward trend across the released checkpoints. They do not establish that parameter scale has no effect: each model supplies its own effect mask and nearest-neighbour set, and the candidates are not exchangeable interventions. A preregistered shared physical effect mask with fixed negative indices is therefore required for a controlled scaling claim. Nor does this section establish that weak action discrimination causes planning failure. Sections 5 and 6 instead test the planner–cost interface directly.

4 The wall, at upstream parity

We match the upstream Metaworld closed-loop evaluation read off its shipped configuration: goal = the scripted expert’s final frame; CEM with horizon 6, 300 samples, 15 iterations, executing 3 model-steps per replan, ≤ 100 env steps; success = the simulator’s flag at episode end. Reaching a faithful protocol required fixing three environment-side reproduction bugs, documented with verification probes in App. E

Table 2: Model-native DROID action-discriminability diagnostic across four released checkpoints. *Top*: effect-conditioned CRA against nearest-neighbour distractors (nominal exchangeable-candidate baseline $1/17 \approx 0.059$): all four models fall in a narrow range in the action-critical regimes (0.031–0.077). Because the effect mask and nearest neighbours are defined in each model’s own latent geometry, this is an observational comparison rather than a controlled scaling law. *Bottom*: $\text{BB}_{\text{boundary}} > 0$ in every cell (boundary-blind), pre-grasp locus in 3/4. A positive control rules out metric saturation: on toy fully-actuated datasets the same diagnostic returns 0.66–1.0.

	DINO-WM	JEPA-WM	V-JEPA2-AC	V-JEPA2-AC-OSS
regime	22M	300M	1B	1B
<i>effect-CRA, nearest-neighbour (chance ≈ 0.059)</i>				
pre-grasp	0.068	0.077	0.061	0.069
gripper actuation	0.031	0.059	0.049	0.031
contact manipulation	0.055	0.045	0.035	0.033
$\text{BB}_{\text{boundary}} (> 0 = \text{boundary-blind})$				
pre-grasp	1.916	1.205	1.933	1.560
free space	0.963	1.103	1.010	0.819
gripper actuation	1.088	1.057	0.643	0.828
contact manipulation	0.421	0.897	0.390	0.595

(after the fixes, rendered observations are on the training manifold: one-step prediction error $1.4\times$ dataset level, latent nearest-neighbour ratio 0.97 (a rendered frame’s distance to its nearest dataset latent, relative to the dataset’s own nearest-neighbour distance; ≈ 1 means indistinguishable from training frames); physics identical, 1.5 mm replay error).

Table 3: Closed-loop Metaworld, 16 paired episodes per task (paired env initialisation and CEM noise; episode-end judging). The harness recovers a Reach rate consistent with the published baseline; contact collapses with the diagnostic signature. “grounded” = the L_2 cost augmented with a 0.5M object-dynamics channel $h(z, a) \rightarrow \Delta_{\text{obj}}$ (App. C).

task	arm	success	final state-dist	final ee-dist
mw-reach	L_2	6/16 (37.5%)	0.332	0.022
	grounded	8/16 (50.0%)	0.412	0.031
mw-push	L_2	0/16	0.624	0.038
	grounded	0/16	0.527	0.028
mw-pick-place	L_2	0/16	0.578	0.035
	grounded	0/16	0.496	0.021

Three readings. (i) Where the diagnostic is clean (free space), the harness is consistent with the published baseline: Reach 6/16, i.e. 37.5% with CI [18.5, 61.4]%, against the reported $44.8 \pm 8.9\%$ — the published value falls inside our interval, though our point estimate sits below its mean; we read this as compatibility from one released checkpoint over 16 episodes, not a seed-matched replication (cf. limitations, Sec. 8). The harness is healthy, the contact failure is real. (ii) The failure signature is specific: final end-effector 2–4 cm from its goal pose while the object never moves — the closed-loop face of the smeared contact bifurcation, matching the upstream reports (“hallucinates grasping”; sub-goal images required). (iii) A grounded auxiliary cost (a 0.5M object-dynamics channel $h(z, a) \rightarrow \Delta_{\text{obj}}$ whose construction and metric-level gains we defer to App. C) improves the cost surface with a CI-clean paired gain (final state-dist +0.089 [+0.022, +0.162] pooled contact) — *and flips zero successes*. That last observation is the loose thread this paper pulls: if the model is the problem, why does repairing the model’s outputs not repair planning?

5 The oracle ladder: the wall is the cost

We isolate which component breaks contact planning by swapping exactly one rung at a time, holding the planner fixed (CEM 100×6 per replan, $H=6$, strict episode-end success, 16 episodes, `dino_wm_metaworld`). This budget is smaller than the parity harness of Sec. 4 (300×15) for compute reasons, but it is *not* the bottleneck: Sec. 6.2 sweeps the CEM budget up to 24 iterations and 300 samples and the contact rungs never improve, while the same budget solves the task when supplied the informative true-state cost. The key device is the *latent oracle*: every CEM candidate is rolled in the *simulator*, rendered, and encoded by the real frozen encoder — perfect dynamics through the true representation, learned predictor removed. Any rung that fails under the latent oracle cannot attribute that failure to learned dynamics; within this harness the remaining tested bottleneck is the cost and its interface with search.

Table 4: The oracle ladder (Metaworld push / pick-and-place, strict episode-end success, 16 episodes per rung except the object-readout and learned-metric rungs, run at 8). Same planner, same budget, same success radii on every rung; each rung changes one component. Perfect dynamics + the true-state cost solves the task; perfect dynamics + *any* frozen-encoder cost does not.

rung	dynamics	cost	readout	push	pick
state oracle	perfect	true state ($\ \text{obj}-g\ + 0.5\ \text{hand}-\text{obj}\ $)	sim	16/16	11/16
latent oracle	perfect	latent L_2	—	0/16	0/16
latent oracle	perfect	object readout ($\ \rho(z)-\rho(z_g)\ $)	probe	0/8	0/8
latent oracle	perfect	learned metric $d_\theta(z, z_g)$	—	0/8	0/8
latent oracle	perfect	<i>true-state cost form</i> , probe readouts	probe	2/16	0/16
latent oracle	perfect	same, <i>off-policy-hardened</i> probes	probe	1/16	0/16
world model	learned F	latent L_2 (upstream)	—	0/16	0/16

Read from the bottom up, the ladder rules out, in order: the **predictor** (perfect dynamics changes nothing: 0/16 either way); the **planner and budget** (the same CEM solves 16/16 given the true-state cost); the **cost formula** (the probe-based rung copies the successful cost’s exact functional form, including the hand-approach shaping term, and still collapses $16 \rightarrow 2$); and **static readout precision** (a spatial object readout ρ (App. D) decodes the object to 92% within 5 cm on held-out frames — the encoder *carries* the object position within the success radius).

The decisive rung: a real readout repair transfers nothing. One hypothesis survives the ladder: CEM scores *off-policy* frames (latents of arbitrary candidate rollouts), where an expert-trained probe was never fit. This is true — the expert probe degrades to 78% within 5 cm on random off-policy frames — and it is fixable: retraining the probes with off-policy frames mixed in restores **91.5%** within 5 cm (push 92.9%, median 2.0 cm) *on the same frame distribution the planner scores*. The re-gate: push 1/16. A measured, targeted repair of the hypothesized binding constraint — readout precision on the planner’s own query distribution — transferred *zero* planning success. Off-policy readout precision was never the binding constraint. Something about the *interaction between the optimizer and the cost* is.

6 Search selects residual cost-error pockets

6.1 Direct evidence: CEM converges onto the cost’s error pockets

The reconciliation of Sec. 5’s paradox is that CEM does not sample off-policy frames *uniformly* — it searches for the cost *minimum* over 100×6 candidates per replan, so any residual-error pocket in the cost is found and converged upon, precisely *because* it scores too low. We test this by decoding the probe on the CEM *elite* population itself — the frames the planner actually trusts — mined from the same hardened-probe runs:

The same probe that reads 2 cm on random frames reads 7–8 cm — beyond the 5 cm success radius — on the frames its own planner selected. The closed-loop consequence is visible in the executed episodes: the cost reports near its minimum (“object at goal”) while the true object sits 13–30 cm away. This is reward

Table 5: Object decode error of the *off-policy-hardened* probe on the CEM elite distribution (13,440 elites; final iteration = the converged population), vs. the same probe on random off-policy frames. The probe is accurate where CEM does not look and inaccurate exactly where it converges.

frame distribution	median err	within 5 cm	within 7 cm
random off-policy (reference)	2.0 cm	91.5%	—
all CEM elites	7.0 cm	24.0%	50.5%
final-iteration elites	7.3 cm	19.2%	46.0%
final-iteration, push	7.7 cm	15.3%	38.9%

hacking in the strict sense (Skalse et al., 2022): the optimizer achieves a low proxy value by exploiting the proxy’s misspecification, not by achieving the goal.

The exploitation is on ground truth, not the probe. Because ρ is itself OOD-sensitive, we verify the *same* elites against simulator state, which never passes through the probe: on the committed (final-iteration) elites the *true* object sits a median 24.9cm from goal (19.9cm on push, 28.0cm on pick), and only 1.7% fall within the 5cm success radius (push 3.4%, pick 0.0%). The planner’s trusted candidates are genuinely far from the goal — the probe collapse in Table 5 reflects real exploitation, not readout noise, and the ground-truth elite quality (1.7% within 5cm) is in fact *worse* than the probe reports (19.2%), so the probe number understates the wall (results/exploit_simstate_crosscheck.csv).

6.2 Search-pressure curves

The framing makes a falsifiable prediction the previous sections do not test: exploitation should be *monotone in optimization pressure*. Holding everything fixed and sweeping the CEM budget, an informative true-state cost should convert extra search into better true outcomes; an exploitable cost should convert it into deeper pockets — proxy improving, truth flat or worse, in direct analogy to the reward-model overoptimization curves of Gao et al. (2023). We test this on the latent oracle (perfect dynamics; only the cost is learned, so any proxy/truth divergence is attributable to the cost under optimization, not to predictor error), sweeping CEM iterations $\in \{2, 6, 12, 24\}$ over $n=16$ paired episodes per cell, across cost $\times$ task arms ($\{\text{push, pick-and-place, reach}\} \times \{\text{probe-based state cost, latent } L_2\}$) of which three isolate the three regimes of interest: exploitable, informative positive-control, exploitable, and no-signal.

The three predictions hold (Fig. 1). **Exploitable cost** (push, probe-based state cost): within a single search at budget 24, the believed-best candidate’s proxy falls monotonically (−18%) while its true object→goal distance does not improve (−2%), and the probe’s decode error on the elite population rises 6.6 cm [5.7, 7.4] → 8.3 cm [7.1, 9.3] (+25%). We state the evidentiary status of that last quantity precisely: the first/last-iteration decode CIs *overlap* at $n=16$, so the deepening pocket is a point-estimate trend, not a CI-separated effect — but it is a trend that appears in *both* exploitable arms (pick-and-place $\times$ probe-cost: 6.7 → 8.1 cm, +21%) and in the informative reach control (reach $\times L_2$: +1%), and it sits on top of the CI-clean across-budget facts that follow. Across budgets the true push object distance does not improve with search (med $\approx 0.18\text{--}0.21$ m at all of $\{2, 6, 12, 24\}$ iterations; episode success 0–2/16, every CI containing the frozen band). **Informative positive-control cost** (reach, L_2): here more search *does* convert to success (5/16 → 16/16 → 16/16 → 16/16 at 2/6/12/24 iterations), confirming the sweep detects “search helps” when the cost is aligned. **No-signal cost** (push, L_2): the proxy falls (−35%) but the true object distance is *exactly* flat (0.239m median at every budget) — L_2 -in-latent has no minimum at task success, so optimizing it is inert; this distinguishes an *exploitable* cost (signal present, worst-case reachable) from a merely *uninformative* one. Pick-and-place corroborates all three modes (probe-cost 0–1/16 with the same rising-decode signature; L_2 flat at 0.286 m). A second pressure axis — CEM population $\in \{50, 100, 300\}$ at fixed 12 iterations ($n=8$ episodes) — corroborates: enlarging the candidate pool does not improve the true push outcome either (probe-cost median 0.233/0.207/0.252 m; L_2 flat at 0.252 m).

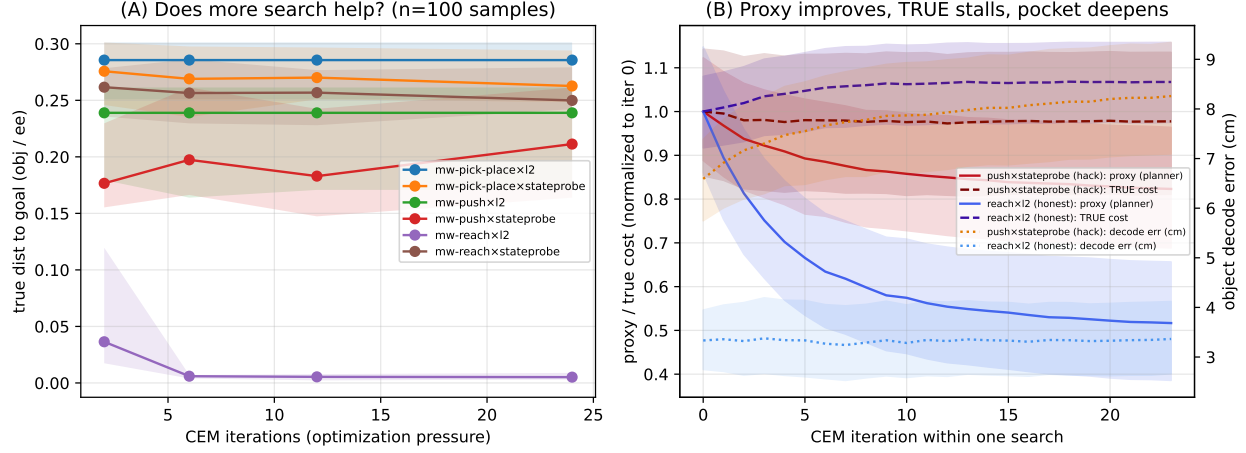


Figure 1: **Search pressure on latent planning costs** ($n=16$ paired episodes per cell; shaded bands are episode-clustered bootstrap 95% CIs). (A) True distance to goal vs. CEM iterations. reach $\times L_2$ (informative cost) converts more search into success ($5/16 \rightarrow 16/16$); push/pick $\times$ probe-cost (exploitable) and push/pick $\times L_2$ (no-signal) do not improve with budget. (B) Within one search at the largest budget, normalized to the first CEM iteration: for push $\times$ probe-cost the proxy the planner minimizes falls (-18%) while the true state cost stalls (-2%) and the object-decode error on the committed elites rises ($6.6 \rightarrow 8.3$ cm, $+25\%$; pick $6.7 \rightarrow 8.1$ cm, $+21\%$) — the planner digs into deeper readout-error pockets. The decode rise is a point-estimate trend (first/last CIs overlap at this n); the informative reach cost (reach $\times L_2$) shows no such trend ($+1\%$).

6.3 Tested mitigation arms do not cross contact

If the diagnosis is right, repairs that improve the cost’s *average* quality without removing its *worst-case* pockets should not cross the wall — the optimizer finds whatever pockets remain. We ran the standard playbook, each arm under the latent oracle (perfect dynamics), push held-out, strict success:

Table 6: Mitigation ladder (latent oracle, push; 16 episodes per arm except the learned-metric arm at 8). Every intervention that operates on or above a frozen-base representation stays inside the frozen baseline’s 0–2/16 band — including those that push readout accuracy to near-ceiling. The true-state cost row shows the same planner and budget solve the task when the cost has no pockets.

intervention	grounding-level metric	push
latent L_2 (frozen null)	—	0/16
probe-based state cost	static decode 92% <5 cm	2/16
off-policy-hardened probes	off-policy 91.5% <5 cm	1/16
learned latent metric d_θ	—	0/8
relearned repr. adapter φ (v2)	held-out decode med. 6.9 cm	1/16
encoder-LoRA + φ (5 seeds)	decode 94–100% <5 cm	{5,0,2,1,1}/16; mean 1.8, CI [0, 3.7]
+ ensemble disagreement, $\lambda \in \{0, \frac{1}{2}, 1, 2\}$	—	{2,1,2,1}/16
true-state cost (no pockets)	—	16/16

Three of these deserve comment. **Encoder fine-tuning** (LoRA on the DINOv2 trunk, action-conditioned grounding objective, 5 seeds): grounding reaches 94–100%, the five-seed planning mean is 1.8/16 with CI [0, 3.7] — statistically inside the frozen band; the single 5/16 seed is high-variance noise, not a crossing (its own 95% Wilson interval, [14, 56]%, overlaps every other arm’s; and every frozen-base arm’s Wilson upper bound is $\leq 3.1/16$ for the $n=16$ rungs — results/ladder_mitigation_cis.csv). This is the strongest form of the paper’s recurring scoped null: the tested grounding objective improves its target metric without producing reliable planning success. **Ensemble disagreement** (cost = $\text{mean}_k d_k^2 + \lambda \text{Var}_k[d_k]$ over 5 inde-

pendently fine-tuned seeds) is the standard pessimism remedy for model exploitation (Janner et al., 2019; Coste et al., 2024); it fails here ($\{2, 1, 2, 1\}/16$ across λ) for an instructive reason: the seeds share the frozen pre-trained base, so they share the blind spot, and disagreement stays flat exactly on the exploited pockets — the same failure mode reported for reward-model ensembles sharing a pre-trained trunk (Eisenstein et al., 2024). **The state-oracle row** is the positive control: the task and planner budget permit success when supplied true state, but it does not identify a unique defect shared by every failed learned cost.

Summary of the mechanism section. A frozen-base learned cost, under the tested search pressure, exhibits the signature of selection into cost-error pockets: CEM selects low-proxy candidates whose simulator-verified outcomes remain poor. The tested readout, grounding, metric, and disagreement repairs do not remove that failure. This is a finite scoped intervention grid, not evidence that every post-hoc cost must fail. Two caveats bound the strength of these nulls. First, statistical: at $n=16$ episodes a 0/16 arm’s Wilson interval reaches 19.4%, so each null rules out mitigations that would lift success above $\sim 20\%$, not arbitrarily small gains — the claim is “no crossing,” not “exactly zero.” Second, multiplicity: the mitigation grid spans many small- n cells, which is precisely why we read the occasional 2/16 (and the single 5/16 seed) as band noise rather than signal — the same logic applies symmetrically and we do not interpret any individual +2/16 cell as a partial success.

7 A candidate predictor intervention: counterfactual training

The ladder bypassed the predictor (perfect dynamics); the deployed stack still uses one. Independently of the cost-side wall, the diagnostic of Sec. 3 identifies a training-target defect in the predictor: under a teacher-forced objective, the target latent already contains the action’s effect, so the objective admits solutions in which the predictor matches targets without using the action channel — a mechanism independently confirmed in a toy setting by Radha and Goktas (2026). The corrective is to make the objective *counterfactual over actions*: the prediction under the factual action must beat predictions under distractor actions, not just match its target.

We instantiate this as an InfoNCE-over-actions objective on the frozen DROID world model (`dino_wm_droid`): a predictor-LoRA of 0.469M parameters (encoder untouched, no object ground truth), trained so that $F(z_t, a_t)$ scores higher against z_{t+1} than $F(z_t, a_k^-)$ for executed-action distractors a_k^- , with a stop-gradient on the target. Concretely, writing $d(a) = \|F(z_t, a) - \text{sg}(z_{t+1})\|^2$ for the predictor’s latent error under action a (sg the stop-gradient) and $\{a_k^-\}_{k=1}^K$ for in-batch permuted (executed) distractor actions, the objective is the action-softmax cross-entropy

$$\mathcal{L}_{\text{cf}} = -\log \frac{\exp(-d(a_t)/\tau)}{\exp(-d(a_t)/\tau) + \sum_{k=1}^K \exp(-d(a_k^-)/\tau)}, \quad (3)$$

with temperature $\tau = \text{sg}(\overline{d(a_t)})$ auto-scaled to the batch-mean factual distance, so the logits stay $O(1)$ regardless of the model’s absolute latent scale (no per-model tuning). The stop-gradient on z_{t+1} is essential: without it the objective is minimized by collapsing the *target* rather than by grounding the *prediction*. An earlier version of this draft reported large gains from this objective on a DROID planning probe. That evaluation was invalid as a held-out claim: although training used trajectory-level train/validation partitions, the downstream planning probe could draw anchors and hard negatives from the complete latent cache, including training trajectories. We therefore remove those numerical results and all method claims based on them.

The corrected protocol persists an immutable 70/15/15 trajectory manifest, embeds its membership and hash in each checkpoint, selects checkpoints on validation only, and restricts both planning anchors and hard-negative pools to the untouched test trajectories. Four training seeds on DINO-WM and JEPA-WM are being rerun under this protocol. Until those runs complete, the objective is a pre-specified candidate intervention rather than evidence or a contribution of this paper. The oracle ladder also implies that even a genuine predictor-side gain would not, by itself, establish a remedy for the cost-side planning failure: learned dynamics have already been removed in the latent-oracle arms.

8 Discussion, scope, and limitations

Elite-conditioned audits. Average held-out metrics need not describe the candidates selected by search. Let c be the cost, P the planner, $\mathcal{E}$ its converged elites, and $\mathcal{D}_{\text{off}}$ an off-policy reference set. For a readout with ground-truth labels, we report the *elite readout shift*

$$G_{\text{readout}}(c, P) = \mathbb{E}_{z \sim \mathcal{E}}[\text{err}_\rho(z)] - \mathbb{E}_{z \sim \mathcal{D}_{\text{off}}}[\text{err}_\rho(z)], \quad (4)$$

where the second error term is also err_ρ (the notation is kept explicit to distinguish the learned readout from the simulator verifier). We separately report the simulator-verified elite outcome and the per-replan proxy-truth change. These quantities answer different questions: a large G_{readout} detects selection-induced readout shift; a poor true elite outcome detects an uninformative or misleading cost; and proxy improving while truth worsens is direct evidence of exploitation. A small readout shift alone is not evidence that a cost is honest—a no-signal cost can be stably wrong.

In our probe-cost arms the same readout moves from 2.0 cm error off-policy to 7–8 cm on CEM elites (91.5% $\rightarrow$ 24% within 5 cm), while simulator state confirms that the selected object positions remain far from goal. By contrast, some latent- L_2 arms have only a small readout shift but also leave the true object 22–27 cm away; we classify these as no-signal, not honest. The current ladder is too small and success-flat for claiming that any single gap is a calibrated predictor of planning success. We therefore present the three-part audit as a mechanism diagnostic and retain success as the endpoint.

What follows for method design. The results sort the design space into three responses. (i) *Harden the cost*: our evidence says this loses on frozen-base representations — average metrics improve, worst-case pockets remain, the optimizer finds them (Table 6). (ii) *Fix the training objective*: the counterfactual objective in Sec. 7 is a candidate predictor-side intervention whose held-out rerun is pending; encoder-level alternatives remain open. (iii) *Remove the adversary*: if test-time search is what weaponizes cost error, amortizing planning into a policy or inverse-dynamics controller (Nguyen et al., 2026), or anchoring the search to demonstration priors (Gao et al., 2026), removes the optimization pressure rather than the pockets. We tested this directly (Sec. 8): a goal-conditioned inverse-dynamics controller built from the same frozen representation and readout, run with *no* search, and the same controller with search re-added in increasing doses. The result is a null and it sharpens the paper’s claim rather than softening it — see below.

Removing search alone is insufficient in this controller. Under the latent oracle on push, the search-free amortized controller solves 0/8, and every re-added search dose (CEM iterations $\{0, 2, 6, 12, 24\}$ seeded from the controller) also solves 0/8; in all arms the object never leaves its start (med $\|obj - goal\| \approx 0.25$ m, Fig. 2A). Removing the optimizer does not cross the wall. Two things are nonetheless confirmed. On the mechanism side, the per-replan seed-vs-chosen comparison shows search “improves” the proxy over the seed on 98–100% of replans while *corrupting* the true state (proxy down, truth up) on a fraction that rises monotonically with dose ($0 \rightarrow 7\%$, Fig. 2B) — the adversary is real and dose-dependent even here. But it is not the only observed failure: this particular controller also fails without an optimizer. That null does not identify whether representation information, controller training, coverage, or control architecture is the cause. The supported synthesis is narrower: the tested search/cost interface is exploitable, and neither the tested cost repairs nor this amortized controller crosses contact on its own. Grounding metrics are therefore insufficient as success certificates; their necessity remains unestablished.

Scope and honest limits. The exploitation mechanism is established on Metaworld, where simulator state lets us verify every claim against ground truth; on DROID the current evidence is an observational action-discriminability diagnostic. Its shared-mask scaling rerun and the held-out predictor intervention are pending, and closed-loop exploitation curves on real robots remain future work. The mitigation table covers the levers available on frozen pretrained checkpoints (readouts, metrics, adapters, LoRA fine-tuning, ensembles); full encoder retraining with planner-aware objectives is beyond our compute and remains open — our claim is about the *post-hoc* regime. The boundary-regime labels use an object-displacement proxy (no MuJoCo contact ground truth in the released data); mw-door-close is excluded as a proxy anomaly. The Reach parity check compares one released checkpoint over 16 episodes against a published 3-seed $\times$ 96-episode number — we read it as “reproduces, inside the CI”, not a seed-matched replication. Finally,

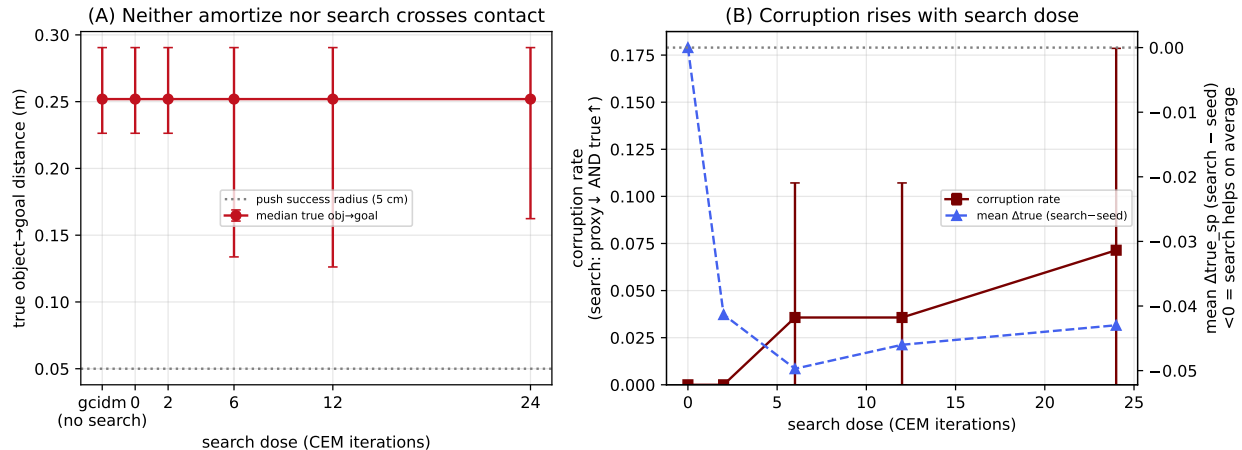


Figure 2: **Removing the search adversary does not cross contact.** (A) True object→goal distance for the pure amortized controller (“gcidm”, no search) and for search re-added in doses; all sit at ≈ 0.25 m, far above the 5 cm success radius. (B) Per-replan corruption rate (search lowers the proxy *and* raises the true cost) rises with search dose ($0 \rightarrow 7\%$), while the mean true-cost change from search stays negative (on average search still helps the *hand*) — the adversary is real but small, and the object never moves regardless.

an information-sufficiency question remains open beneath everything: vision-only latents may lack contact state outright (Zhang et al., 2026); our state-oracle row shows the *task* is solvable from privileged state, but whether any vision-only cost can be made pocket-free is not settled by this paper.

9 Conclusion

On the tested MetaWorld contact tasks, latent-space planning failure persists when learned dynamics are replaced by simulator-perfect rollouts, while the same planner succeeds with privileged true-state cost. CEM elites reveal why the tested learned costs fail: proxy scores improve on candidates whose simulator-verified outcomes remain poor, and learned readouts degrade under selection. The result is not a universal impossibility claim about latent costs, scale, or grounding. It is a controlled warning that average prediction and readout metrics do not certify a cost under test-time search. World models intended for planning should therefore report optimizer-conditioned audits and true task outcomes alongside average-case metrics.

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A Regime stratification

Every per-regime table in the paper (Tables 1, 2, and 7) relies on assigning each transition ($s_t \rightarrow s_{t+1}$) to exactly one of four regimes. The assignment is a *priority cascade*: the first matching rule wins, so the regimes are mutually exclusive by construction and the ordering encodes precedence (the gripper-actuation *event* outranks the contact *state* it may create). Thresholds are fixed constants (below), not fit to any outcome.

Metaworld (simulator state). The released HuggingFace dataset ships no MuJoCo contact flags but does ship the full 39-dim Metaworld-v2 observation, from which we recover the end-effector position (`state[0:3]`), gripper aperture (`state[3]`), and primary-object position (`state[4:7]`) exactly. For a transition:

1. **gripper actuation** if $|\Delta \text{grip}| > 0.10$ (the gripper is opening/closing — the grasp/release event itself);
2. else **contact manipulation** if $\|\Delta \text{obj}\| > 5 \text{ mm}$ (the object measurably moved — something pushed it);
3. else **pre-grasp** if $\|\text{ee} - \text{obj}\| < 10 \text{ cm}$ (the hand is close to the object but has not yet moved it);

4. else **free space**.

“Contact” here therefore means *the object displaced*, an object-displacement *proxy* rather than a contact sensor; the caveat is carried on every Metaworld figure, and `mw-door-close` is excluded as a proxy anomaly (its hinge geometry violates the displacement assumption).

DROID (real gripper sensor + latent-change proxy). DROID-wrist has no object pose, so contact is read from the *real gripper sensor* (primary) and the approach cue from the latent-change proxy (secondary):

1. **gripper actuation** if $|\Delta\text{grip}| > 0.2$;
2. else **contact manipulation** if the gripper is held *closed* ($\text{grip}_t > 0.5$ on the $[0, 1]$ open→closed scale) — holding or contacting an object, a sensor signal that (deliberately) does *not* depend on visual motion, so a closed gripper resting on an object counts as contact even when the scene is momentarily static;
3. else (gripper open) **pre-grasp** if the scene is actively changing, $\|\Delta z\| > 1.0 \times \text{median}\|\Delta z\|$ (an object entering/growing in the wrist view; the dataset-median baseline is the same one used for effect-conditioning);
4. else **free space** (gripper open, scene static).

Making contact gripper-primary rather than $\|\Delta z\|$ -gated matters for the raw per-regime counts (the earlier motion-gated rule mislabeled $\sim 54\%$ of closed-gripper *static-holding* steps as free space); it leaves the *headline* numbers unchanged, because those are effect-conditioned ($\|\Delta z\| > \text{median}$) and the relabeled steps are the low- $\|\Delta z\|$ holding steps that effect-conditioning excludes either way.

RoboCasa (grasp flags). RoboCasa exposes an `object_in_hand` flag and proprioception, mapped to the same taxonomy: gripper actuation if $|\Delta\text{grip}| > 0.2$; else contact manipulation if `object_in_hand` at t or $t+1$; else pre-grasp if $\|\text{ee} - \text{obj}\| < 12$ cm (looser than Metaworld for the larger scene scale); else free space.

The three datasets thus share one taxonomy and one cascade ordering, differing only in the contact signal available (object displacement / gripper sensor / grasp flag). Source: `diagnosis/stratification/{metaworld,droid,robocasa}`

Robustness to the threshold choice. Because the pre-grasp locus (Table 1) is read off a fixed 5 mm/10 cm/0.10 cut, we re-ran classification and the hard-nearest-neighbour diagnostic (Sec. 3.1) under six perturbations — object-move $\in \{2.5, 10\}$ mm, pre-grasp distance $\in \{8, 12\}$ cm, gripper-delta $\in \{0.05, 0.20\}$ — against the baseline. Pooled effect-conditioned CRA at pre-grasp moves by at most 0.037 (DINO-WM: $[0.450, 0.488]$; JEPA-WM: $[0.529, 0.558]$) across all seven configurations, and **pre-grasp remains the lowest-scoring regime, below both free space and contact manipulation, in every single configuration for both models** (`results/regime_robustness_summary.csv`, scripts/04+46). The locus of the collapse is a property of the transitions, not an artifact of the specific cut.

B Boundary Blindness: definition

The instrument compares, on a neighbourhood of similar states, how much the *world’s* outcome fans out under nearby actions against how much the *model’s* prediction fans out under the same actions.

Neighbourhood. For each anchor transition i with latent z_i and action a_i , $\mathcal{N}(i)$ is its $M=16$ nearest cached transitions in latent distance, radius-limited with relaxation (the same neighbourhood the hard-negative CRA sampler uses). o_j is the true scalar outcome of neighbour j : object displacement $\|\Delta\text{obj}_j\|$ on Metaworld, the $\|\Delta z_j\|$ proxy on DROID.

Boundary regime. A transition is *boundary* when small action changes produce large true-outcome changes across its neighbourhood:

$$\text{boundary_score}_i = \frac{\text{std}_j o_j}{\text{mean}_j \|a_j - a_i\| + \varepsilon}, \quad (5)$$

thresholded at the 75th population percentile. Free-space drift has large $\|\Delta z\|$ but smooth neighbourhoods and scores low.

The two sensitivities and BB. The world’s local action-sensitivity S_i^{true} is the dispersion of true outcomes across the neighbourhood; the model’s, S_i^{model} , applies each neighbour’s action to the *single* anchor state and measures the RMS spread of $F(z_i, a_j)$ around its centroid. Each is z-scored over the population and

$$\text{BB}_i = \text{relu}(\tilde{S}_i^{\text{true}} - \tilde{S}_i^{\text{model}}), \quad (6)$$

pooled (n_b -weighted) over the boundary mask with trajectory-clustered bootstrap CIs. Because BB references the world, a high value cannot be an artefact of small one-step latent deltas.

C The grounded-channel fix ladder (metric-level repair)

Deferred from Sec. 4: the model-side repair ladder that motivated the oracle ladder. **Nulls:** mixture-density predictor heads (soft-NLL/WTa/hard-EM/supervised, $K \in \{2, 3\}$) beat their $K=1$ NLL controls but do not move $\text{BB}_{\text{boundary}}$ — the bifurcation spans 9.9 L_2 units against a 106 median residual (the geometry under-weights the boundary subspace 10 $\times$), and the boundary’s action-dependence is unlearnable from expert-only data at the head level. Probe-subspace metric re-weighting redistributes BB without reducing it. **The probe chain:** object state is decodable from the frozen latent (V1 err 0.064 vs. sd 0.094), propagates through the predictor for *factual* actions (V2 0.059), but the *counterfactual* action $\rightarrow$ object response is noise (V3 spread-corr +0.035). **The repair:** a 0.5M grounded dynamics channel $h(z_t, a) \rightarrow \widehat{\Delta_{\text{obj}}}$ (mean/max-pooled tokens + action MLP), trained on cached latents with everything frozen, restores counterfactual tracking (0.035 $\rightarrow$ 0.682 DINO-WM; 0.702 JEPA-WM), halves boundary BB on both families (1.323 $\rightarrow$ 0.660; 1.280 $\rightarrow$ 0.620), and improves the closed-loop cost surface (Table 3) — while flipping zero successes, the observation Sec. 5 explains. Honest negative: the recipe does not transfer to DROID’s noisy whole-state proxy label at 2.1k transitions (rank-corr +0.708 but spread magnitude collapsed; BB unchanged) — the principle needs a learnable grounding label.

D Probe, verifier, and mitigation-arm implementations

Every learned rung of the oracle ladder (Table 4) and mitigation table (Table 6) is small and trained on the cached latents with the encoder and predictor frozen. We give each arm’s spec so the “all repairs fail” claim can be audited.

Object probes (the readout ρ). Two readouts decode object xyz from a single frozen latent, trained supervised on cached (latent, simulator object-position) pairs. The *pooled* probe is a mean+max-pooled-token MLP ($2D \rightarrow 512 \rightarrow 256 \rightarrow 3$, SiLU). The *spatial* probe prepends a learned CLS query to the patch tokens and routes it through a 2-layer, 6-head Transformer encoder before a linear head; being permutation-aware it can attend to *which* patch lit up, so it upper-bounds the spatial information the latent carries — it is the readout behind the 92%-within-5 cm decode in Table 4. The *expert* probe trains on expert-trajectory frames only; the *off-policy-hardened* probe mixes in latents of arbitrary candidate rollouts, which restores 91.5% within 5 cm on that off-policy distribution.

On using ρ as the verifier. The exploitation evidence (Table 5) decodes ρ on the CEM elites. Because ρ is itself OOD-sensitive — that sensitivity is precisely what is under test — we do not rest the claim on ρ alone: on the *executed* Metaworld episodes the simulator’s true object sits 13–30 cm from goal while the cost reads near its minimum (Sec. 6.1), a ground-truth corroboration that does not pass through ρ . On real-robot data ρ (or a held-out verifier) is the only available instrument, which is why the evaluation protocol (Sec. 8) is stated for a generic verifier rather than privileged state. A tighter closed-loop version — decoding elites against simulator state directly, not ρ — is the natural next confirmation and is left as future work.

Grounded object-dynamics channel h (Sec. 4, App. C). $h(z_t, a) \rightarrow \widehat{\Delta\text{obj}}$ is a mean+max-pooled-token MLP with an action-embedding branch ($2D+64 \rightarrow 512 \rightarrow 256 \rightarrow 3$); the grounded cost integrates it along the rollout, $\widehat{\text{obj}}_H = \rho(z_0) + \sum_t h(\hat{z}_t, a_t)$, and scores $\text{MSE}(z_H, z_g)/s_z^2 + \beta^2 \|\widehat{\text{obj}}_H - \rho(z_g)\|^2/s_g^2$, the two terms rescaled by dataset scales s_z, s_g because they span $\sim 98\text{k}$ vs. 3 dimensions (per-dim MSE would silently collapse the blend to object-only).

Learned latent metric d_θ (Tables 4, 6). A goal-conditioned quasimetric over pooled tokens, $d_\theta(z, z_g) = \|u_s(z) - u_s(z_g)\|_2 + \sum_i \text{relu}(u_a(z)_i - u_a(z_g)_i)$ — a symmetric part plus a directed residual (MRN-style), so $d_\theta(z_g, z_g) = 0$ is the unique minimum. Trained *without* object labels on within-trajectory temporal order ($d_\theta(z_t, z_{t'}) \approx t' - t$ for $t < t'$, cross-trajectory pairs pushed large), so the recipe is dataset-agnostic.

Relearned representation adapter φ (Table 6). $\varphi = A(z)$ is a small head over the frozen patch tokens, trained with four losses so that plain L_2 in φ -space is a harder-to-exploit cost: (i) *grounding* — the first obj_dim coordinates regress true object xyz; (ii) *counterfactual margin* — on hard-effect negatives (same z_t , different action, genuinely different true outcome) the two object estimates must be separated by at least their true gap, a margin constraint that targets residual-error *pockets* rather than average accuracy; (iii) *temporal ranking* confined to the non-object subspace; (iv) *adversarial* — previously-mined CEM elite frames are margin-pushed from their episode goal in the object subspace. This is the strongest post-hoc cost we built, and it still gates push $1/16$.

Encoder-LoRA (Table 6). Zero-initialized LoRA adapters ($B=0$, so the injected model starts *exactly* at the frozen encoder) inserted into the encoder’s transformer blocks and trained with the φ losses; unlike φ — a function of the frozen z , so it can only read around entangled latents — this can separate them at the source. The adapters are toggleable, so one injected model serves both the frozen-baseline and LoRA arms; we train five seeds.

Ensemble disagreement (Table 6). Five independently fine-tuned encoder-LoRA seeds, with cost $\text{mean}_k d_k^2 + \lambda \text{Var}_k[d_k]$ — the standard pessimism penalty — swept over $\lambda \in \{0, \frac{1}{2}, 1, 2\}$.

E Reproduction integrity

Closed-loop sim evaluation is fragile, and getting the baseline wrong would invalidate the parity argument. Three environment-side bugs, found and fixed, all model-agnostic: **(1)** Metaworld’s offscreen renderer is constructed in the env constructor, so camera/size assignments afterwards were silently ignored (default camera at 480 px vs. the trained **corner2** at 224 px; symptom: one-step prediction error $8.5\times$ dataset level). We re-create the renderer explicitly, mirroring the upstream wrapper. **(2)** The released training frames are vertically flipped relative to today’s **corner2** render; pixel-calibration against dataset initial frames scores the flipped wrapper-camera render at MSE 71.6 vs. ≥ 3000 for every unflipped candidate. **(3)** The success flag fires at a 5 cm radius but the scripted expert refines to ~ 1 cm; taking the goal frame at first-success rather than the expert’s final frame parked the planner just outside the radius. **Verification:** after the fixes, one-step prediction error is $1.4\times$ dataset level, latent NN ratio 0.97, and replaying dataset actions reproduces the proprioceptive trajectory to 1.5 mm median end-effector error — only the visuals had drifted, never the physics. The reach baseline then recovers from near-zero to 37.5% (Table 3), inside the published CI.

F AUG and ECS: the raw-gap corroboration

CRA leads the main text because, as a ranking, it is cross-model comparable. The two raw-gap metrics corroborate it: AUG (prediction-error gap between permuted and factual actions; ≈ 0 iff actions ignored) and ECS (AUG on effect-bearing transitions), both on the hard nearest-neighbour family.

Under the model-native DROID protocol, the effect-conditioned action gap remains small across the released checkpoints. This observational result is compatible with weak action use, but it does not by itself identify

Table 7: AUG / ECS, hard nearest-neighbour family, per regime (n_{eff} -weighted pools; dash = no effect-bearing transitions). Raw latent gaps are *not* comparable across models (latent scales differ); the cross-model statement lives in CRA (Tables 1, 2).

model	free space	pre-grasp	gripper	contact
<i>DROID</i> (real robot; AUG / ECS)				
DINO-WM (22M)	0.038/−	0.011/0.0105	0.038/0.0645	0.070/0.0094
JEPA-WM (300M)	0.000/−	0.000/0.0002	0.001/0.0004	0.001/−0.0011
V-JEPA-2-AC (1B)	0.006/−	0.002/0.0016	0.003/0.0026	0.003/−0.0059
V-JEPA-2-AC-OSS (1B)	0.004/−	0.001/0.0010	0.001/0.0005	0.003/−0.0030
<i>Metaworld</i> (sim; AUG / ECS; pooled over 12 tasks)				
DINO-WM (22M)	0.088/0.1007	0.051/0.0997	−/−	0.081/0.0893
JEPA-WM (300M)	0.089/0.1020	0.058/0.1114	−/−	0.095/0.1052

causal action-ignoring or constitute a controlled scale law. Source: `results/{droid,metaworld}_diagnostic.csv`, `hard_nn` rows.

G Full per-task tables

Per-task Boundary-Blindness dumps, per-arm mitigation episode tables (phase C/D/F/G logs), the overoptimization-sweep curve tables, and the historical Phase-H seed-sweep outputs are provided in machine-readable form under `diagnosis/results/`; the latter are invalid as held-out evidence and retained only for provenance (see `diagnosis/docs/CLAIMS_EVIDENCE.md` for the row-to-table mapping).